

DIPEPTIVEN[®]

Active substance: N(2)-L-alanyl-L-glutamine

QUALITATIVE AND QUANTITATIVE COMPOSITION

100 ml contains:

20 g N(2)-L-alanyl-L-glutamine (= 8.20 g L-alanine, 13.46 g L-glutamine)

Water for injections

theoretical osmolarity : 921 mosmol/l

titration acidity : 90 - 105 mmol NaOH/l

pH value : 5.4 - 6.0

PHARMACEUTICAL FORM

Concentrate for solution for infusion.

THERAPEUTIC INDICATIONS

Dipeptiven[®] is indicated as part of an intravenous parenteral nutrition regimen as a supplement to amino acid solutions or an amino acid containing infusion regimen, e.g. in patients in hypercatabolic and/or hypermetabolic states.

POSODOLOGY AND METHOD OF ADMINISTRATION

Adults:

Dosage depends on the severity of the catabolic state and on amino acid requirement. A maximum daily dosage of 2 g amino acids per kg body weight should not be exceeded in parenteral nutrition. The supply of alanine and glutamine via Dipeptiven[®] should be taken into consideration in the calculation. The proportion of amino acid supplied through Dipeptiven[®] should not exceed approximately 30% of the total amino acid supply.

Daily dose

1.5-2.5 ml of Dipeptiven[®] per kg body weight (equivalent to 0.3 - 0.5 g N (2)-L-alanyl-L-glutamine per kg body weight). This equates to 100 to 175 ml Dipeptiven[®] for a patient of 70 kg body weight.

Maximum daily dose: 2.5 ml equivalent to 0.5 g N (2)-L-alanyl-L-glutamine of Dipeptiven[®] per kg body weight.

The maximum daily dose of 0.5 g of N (2)-L-alanyl-L-glutamine per kg body weight should be administered in combination with a compatible amino acid solution providing at least 1 g amino acids per kg body weight per day. This results in a daily dose of at least 1.5 g amino acids per kg body weight per day.

The following adjustments are examples for the supply of Dipeptiven[®] with other amino acids through the carrier solution:

- Amino acid requirement 1.2 g per kg body weight per day: 0.8 g amino acids + 0.4 g N(2)-L-alanyl-L-glutamine per kg body weight.
- Amino acid requirement 1.5 g per kg body weight per day: 1.0 g amino acids + 0.5 g N(2)-L-alanyl-L-glutamine per kg body weight.
- Amino acid requirement 2 g per kg body weight per day: 1.5 g amino acid + 0.5 g N(2)-L-alanyl-L-glutamine per kg body weight.

The rate of infusion depends on that of the carrier solution and should not exceed 0.1g amino acids per kg body weight per hour. Dipeptiven® is an infusion solution concentrate which is not designed for direct administration. It should be mixed with a compatible amino acid carrier solution or an amino acid containing infusion regimen prior to administration. One volume part Dipeptiven® is to be mixed with at least 5 volume parts carrier solution (e.g. 100ml Dipeptiven® + at least 500ml amino acid solution). 3.5% of the active ingredient should be the maximum concentration during therapy. The duration of use should not exceed 3 weeks.

For central venous infusions after addition to a compatible infusion solution: Solutions of mixtures with an osmolarity above 800 mosmol/l should be infused by the central venous route.

CONTRAINDICATIONS

Dipeptiven® should not be administered to patients with severe renal insufficiency (creatinine clearance < 25 ml/minute), severe hepatic insufficiency, circulatory shock, hypoxia, multiple organ failure, severe metabolic acidosis or known hypersensitivity to the active substances or to any of the excipients.

SPECIAL WARNINGS AND PRECAUTIONS FOR USE

For a safe administration the maximum dose of Dipeptiven® should not exceed 2.5 ml (corresponding to 0.5 g N(2)-L-alanyl-L-glutamine) per kg body weight per day (see section 4.2, 4.9 and 5.1).

Dipeptiven® should only be used as part of clinical nutrition, and its dosage is limited by the amount of protein/amino acids provided by nutrition (see section 4.2). Whenever the clinical condition does not allow nutrition (e.g., circulatory shock, hypoxia, unstable critically ill patients, severe metabolic acidosis) Dipeptiven® should not be administered.

Oral/enteral intake of glutamine-supplemented formulas in combination with parenteral nutrition should be taken into consideration for calculation of the prescribed dose of Dipeptiven®.

It is advisable to regularly monitor liver function parameters in patients with compensated hepatic insufficiency.

As there is currently Insufficient data on administration of Dipeptiven® to pregnant women, nursing mothers and children. administration of the preparation. in these patient groups Is not recommended.

Serum electrolytes, serum osmolarity, water balance, acid-base status, creatinine clearance, urea, as well as liver function tests (alkaline phosphatase, ALT, AST), possible symptoms of hyperammonaemia should be controlled.

The enzymes alkaline phosphatase, GPT, GOT, bilirubin level and the acid-base status should be monitored. The choice of a peripheral or central vein depends on the final osmolarity of the mixture. The general accepted limit for peripheral infusion is about 800 mosmol/l but it varies considerably with the age and general condition of the patient and the characteristics of the peripheral veins.

Experience with use of Dipeptiven® for longer periods than nine days is limited.

INTERACTION WITH OTHER MEDICINAL PRODUCTS AND OTHER FORMS OF INTERACTION

No interactions are known to date.

USE DURING PREGNANCY AND LACTATION

Due to lack of experience, Dipeptiven® should not be administered during pregnancy and lactation.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

Not applicable.

UNDESIRABLE EFFECTS

None known when correctly administered.

Reporting of suspected adverse reactions

Reporting suspected adverse reactions after authorisation of the medicinal product is important. It allows continued monitoring of the benefit/risk balance of the medicinal product. Healthcare professionals are asked to report any suspected adverse reactions via the national reporting system.

OVERDOSE

As with other infusion solutions, chills, nausea and vomiting can occur, when the recommended infusion rate of Dipeptiven® is exceeded.

Infusion should be stopped immediately in this case.

Experience from a study in critically ill patients with at least two organ failures at admission, receiving the maximum approved daily intravenous infusion of Dipeptiven® (0.5 g alanyl-glutamine/kg/day) together with a high dose of enteral glutamine (30 g) provided as a mixture of alanyl-glutamine and glycyl-glutamine and without appropriate clinical nutrition, has shown an increase in serious side effects.

PHARMACODYNAMIC PROPERTIES

The dipeptide N(2)-L-alanyl-L-glutamine is endogenously split into the amino acids glutamine and alanine hereby supplying glutamine with infusion solutions for parenteral nutrition. The released amino acids flow as nutrients into their respective body pools and are metabolised according to the needs of the organism. Many disease conditions, in which clinical nutrition is indicated, are accompanied by a glutamine depletion.

In a large multicenter study, critically ill adult patients with at least two organ failures at admission and requiring mechanical ventilation received either supplemental glutamine alone, antioxidants, glutamine and antioxidants, or placebo. In the glutamine groups, patients concomitantly received parenteral and enteral glutamine in their maximally allowed amounts exceeding the recommended dose by two-fold. Overall, mortality for the entire study population at 28 days, the primary endpoint, was not statistically significantly different between the groups. However, 6-month mortality, in a retrospective analysis, was tendentially increased in patients receiving the combined very high total dose of glutamine during unresuscitated shock and renal failure; glutamine and nutrition should not be used in unresuscitated shock accompanied by renal failure (see section 4.9). Under these specific circumstances the patient's ability to metabolize glutamine seems to have been exceeded (see also section 4.4).

PHARMACOKINETIC PROPERTIES

N(2)-L-alanyl-L-glutamine is rapidly split into alanine and glutamine after infusion. In man, half-lives of between 2.4 and 3.8 min (in terminal renal insufficiency 4.2 min) and a plasma clearance of between 1.6 and 2.7 l/min were determined. The disappearance of the dipeptide was accompanied by an equimolar increase of the corresponding free amino acids. Hydrolysis probably takes place exclusively in the extracellular space. Renal elimination of N(2)-L-alanyl-L-glutamine under constant infusion is below 5% and thus the same as that of infused amino acids.

PRECLINICAL SAFETY DATA

Acute and sub chronic toxicity: A matrix of dosage finding tests were conducted on rats and dogs over 1 to 7 days. In the rats, infusion of 50ml/kg body weight of a 10%, 15% 20% and 30% solution of N(2)-L-alanyl-L-glutamine over 4h/day led to tonic spasms, increased respiratory rate and exitus. Infusion of 50ml/kg body weight of 10% solution (5 g N(2)-L-alanyl-L-glutamine/kg body weight) resulted in necrotic areas at the infusion site, reduced

body weight and yellowing of the kidneys in the rats (6h/day) and a temporary increase in heart rate in the dog (8h/day).

Investigations were carried out in dogs (8h/day) and in rats (6h/day) with 0.5 and 1.5 g N(2)-L-alanyl-L-glutamine/kg body weight per day i.v. over 13 weeks and with 4.5 g N(2)-L-alanyl-L-glutamine/kg body weight per day i.v. over 6 weeks. In the dogs, vomiting occurred. With the high dose tonic or tonic-clonic cramps, increased salivation, ataxia, sedation, and lateral position were observed. Mutagenic and tumorigenic potential: In vitro and in vivo test gave no indications of mutagenic potential. Studies investigating the tumorigenic potential were not carried out. Carcinogenic effects are not to be expected.

Reproduction toxicity: In animal trials, no indications of teratogenic or other embryotoxic and peripostnatal injuries could be observed up to a dosage of 1.6 g N(2)-L-alanyl-L-glutamine/kg body weight per day.

Local tolerance: Following repeated i.v. infusion of N(2)-L-alanyl-L-glutamine (5 and 10% solution) over 13 weeks, intolerance reactions occurred at the infusion sites (swellings, discolorations, necroses) in the rats and dogs from 0.5 g/kg body weight onwards. Histopathologically, substance-induced inflammatory reactions with mild to fully developed dermatitis purulent necroticans and osteomalacia of the tail vertebrae, thrombophlebitis and periphlebitis, were observed in the rats. In the dog, perivascular inflammatory reactions and, occasionally, vessel blockage were observed. The tests conducted on the dog on local tolerance after a single, intra-arterial, paravenous and intramuscular administration gave no indications of unusual intolerance reactions with incorrect administration.

LIST OF EXCIPIENTS

Water for injection.

SHELF-LIFE

24 months.

To be used immediately after the bottle is opened.

Dipeptiven® is not to be stored after addition of other components.

SPECIAL PRECAUTIONS FOR STORAGE

Do not store above 25°C.

Store in the original package.

NATURE AND CONTENT OF CONTAINER

50 ml glass bottle

100 ml glass bottle

250 ml glass bottle

Not all pack sizes may be marketed

INSTRUCTIONS FOR USE AND HANDLING

Dipeptiven® is an infusion solution concentrate which is not designed for direct administration. The container and the solution should be inspected visually prior to use. Use only clear, particle-free solution and undamaged container. For single use only.

The addition of the concentrate to the carrier solution prior to application should take place under aseptic conditions.

Thorough mixing and compatibility must be ensured. Unused solution should be disposed off.

Dipeptiven® is infused with the carrier solution. For details see "Posology and method of administration".

MANUFACTURED BY

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MARKETING AUTHORISATION HOLDER/IMPORTER

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